

Listen to the Sword: Using Breathing and Spatial Audio for Wuxia Narratives

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ABSTRACT

Wuxia, a famous Chinese genre combining martial arts and philosophy, has been portrayed in various films, novels, and games. However, modern media, especially gaming, often prioritizes the martial arts action instead of the rich philosophical elements central to Wuxia. Addressing this gap, we developed "Listen to the Sword," a game incorporating novel auditory gameplay for an immersive Wuxia narrative. This game integrates spatial sound effects with a breath detection system, providing players with an experience that mirrors the self-cultivation journey characteristic of traditional Wuxia. In this way, we hope to investigate the feasibility of gamifying complex cultural concepts to facilitate intercultural communication.

CCS CONCEPTS

• **Human-centered computing** → *Interaction design*.

KEYWORDS

Wuxia genre, gamification, breathing-based interaction, spatial audio

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1 INTRODUCTION

Wuxia, a genre of Chinese fiction and film, holds a significant place in Chinese culture, weaving tales of martial arts, honor, and a unique code of chivalry [15]. Rooted in traditional Chinese literature, Wuxia narratives often explore the balance between power and morality, set against the backdrop of ancient dynasties [19]. The genre's narratives typically involve individuals rising above

personal challenges through rigorous training and spiritual development. This focus on self-cultivation aligns with traditional Chinese philosophies, adding layers of cultural depth to Wuxia storytelling [16]. This is why Wuxia is still popular in the Chinese-speaking community, and being integrated into various forms of media, such as games.

In games inspired by Wuxia, the protagonist's self-cultivation, which encompasses martial arts training and spiritual exploration [16, 19], is translated into game mechanics. This sees martial arts techniques represented as in-game skills, while the cultivation process is typically reduced to a system of experience points. However, this can result in the protagonist's internal and spiritual development, primarily conveyed through narrative text and visual elements, not being thoroughly incorporated into the core game mechanics[3].

To explore the potential of gamifying the spiritual exploration and growth inherent in Wuxia cultivation, we designed *Listen to the sword*. Inspired by one of the renowned spiritual states described in Wuxia novels: "Wang Wo" (Selflessness), and the method required to cultivate to these state: "Zhi Guan" (Stop and Observe) and "Tu Na" (Breath Control). We designed a series of game mechanics that require players to actively control their breath and pay attention to the surrounding environmental conditions, including visual and auditory aspects. In addition, we employed Shui Mo (Ink wash painting) style in the game visuals to emphasize Chinese culture.

2 RELATED WORK

2.1 Wuxia in games

Wuxia has been introduced in games since the 1990s [3]. Since Wuxia usually has a detailed description of the world and battle system that aligns with Kungfu, a Role-playing Game (RPG) is the most suitable game type for Wuxia. There are several famous game series about it, such as *Legend of Sword and Fairy*, and *Ho Tu Lo Shu*[3]. *Ho Tu Lo Shu* offers an open-world RPG immersed in a Wuxia-inspired universe that echoes the landscapes and ethos of Chinese Wuxia art fiction, which resonates with traditional Wuxia settings. It also features an intricate Wuxia arts skill tree system, allowing players to acquire a variety of combat styles and techniques. *Qu Zhong Jian* [8], a novel action-music game, integrates the essence of Wuxia arts with rhythm-based gameplay. In *Qu Zhong Jian*, traditional rhythm game mechanics are enhanced by incorporating Wuxia-themed

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combat sequences that align with the musical rhythm, offering an immersive experience.

These games primarily emphasize Wuxia’s visual and audio aspects, in contrast our project seeks to delve deeper into culture and present a balanced representation of both the physical and philosophical dimensions of Wuxia. The game is designed to facilitate an immersive gamified experience in traditional Wuxia arts cultivation, integrating elements of Tao and Zen philosophy. Our objective is to offer players from diverse cultural backgrounds an opportunity to engage in a cross-cultural journey, thereby enriching their understanding and appreciation of traditional Chinese philosophical concepts within a Wuxia context. In addition, *Listen to the Sword* introduces a unique breath control system as an integral part of its gameplay, further enhancing the player’s immersion in the Wuxia-themed environment.

2.2 Breathing based interactions

Breathing is one of the unique bodily signals that human body can voluntarily control, and has been used for HCI in the past [10, 12]. One example of using breathing-based interaction is work of Wald et al. [17], who created an immersive virtual space featuring a continuously rising bubble that expands and contracts in response to the participant’s breath. They discovered that by visualizing breath in this manner, participants could become more immersed in the interactive process, enhancing their autonomy and bodily awareness.

Weerdmeester et al. [18] created a virtual underwater world where participants need to control the character’s movement through diaphragmatic breathing. The deeper and slower the breathing of the participant, the longer the in-game character can act. The participant’s task is to move the character to the journey’s end. In addition, various elements in the world also respond to the participant’s breath; for example, flowers change color intensity with breath, and the rings within the character’s field of view expand and contract with breath.

Compared to previous work applying breath interaction, we attempted to establish a connection between breath interaction and the experience of traditional Chinese culture. Inspired by the “Tu Na” (Exhale and Inhale) often used by Wuxia protagonists during self-cultivation, in *Listen to the Sword*, we correlated the participant’s breathing with the speed of enemy actions. As participants become more proficient in using breathing techniques, they are granted more time to confront enemies. We hope that such a design allows participants to immerse themselves more deeply in the experiential process of Wuxia cultivation.

3 GAME DESIGN

In Chinese Wuxia novels, there’s a recurring theme named “Wang Wo”(Selflessness), where martial artists enter a state of heightened awareness with time slowing down[4]. It may come from an important concept of Zen - Zhi Guan(Stop and Observe), which means suppressing the delusions in the mind arising from mundane thoughts, keeping the mind calm and stable, and concentrating the mind to observe and think[5]. This phenomenon is often depicted in intense combat scenes, allowing the martial artist to perceive and react to fast-moving events with supernatural precision and calmness. This



Figure 1: Gameplay scene: (a) Instruction of level and control; (b) The visual cue of the opponent; (c) Player’s action outcome; (d) Instruction of Box-breathing; (e) Player’s status box.

concept, while intriguing, presents challenges in communication to players unfamiliar with Chinese cultural contexts. *Listen to the Sword* is designed to present a gamified method for this challenge.

3.1 Main Gameplay

Listen to the Sword is a sword fighting game where the core gameplay revolves around a spatial auditory mechanic. The interaction is primarily auditory; players must rely on sound cues to orient themselves and counterattack using precise sword strikes. Players listen for enemy footsteps to determine the direction of incoming threats. An audio cue signals the optimal moment for attack, enhancing player engagement with the game’s auditory focus. The unique spatial auditory gameplay is designed to encourage players to play the game with their eyes closed, which is aligned with “Zhi Guan” – helping to achieve inner peace by giving up sight. The game’s controls are straightforward, with ‘A’ and ‘D’ keys for directional rotation, and “Left Mouse Button” to make sword strokes.

The gameplay hinges significantly on timing and precision. Players’ actions, determined by their keystroke timing, lead to three distinct outcomes: Perfect, Good, or Miss. A ‘Perfect’ hit indicates impeccable timing, resulting in substantial damage to enemies. A ‘Good’ hit, while effectively parry, is less impactful and causes no damage, and a ‘Miss’ represents a timing error, requiring players to swiftly recover to fend off enemy counterattacks. Progression is tied to performance, with five ‘Perfect’ hits advancing the player to the next level, whereas five ‘Misses’ result in a game loss.

Listen to the Sword is structured across three increasingly challenging levels[14]. This design mirrors the concept of self-cultivation approaching Zhi Guan. Additionally, this escalating challenge serves to maintain player engagement over time. The first level serves an instructional purpose, where players engage with visible opponents, allowing them to familiarize themselves with the game’s mechanics. In the second level, the opponents become invisible, and players have to listen to the spatial footsteps to find them, which guides players to only rely on auditory cues. In the third level, the game environment transitions into complete darkness. This design choice aims to encourage a state of tranquility and introspection. This approach is intended to facilitate a deeper connection with the concept of Zhi Guan, as players are guided to concentrate on their inner experiences rather than the external visual environment.

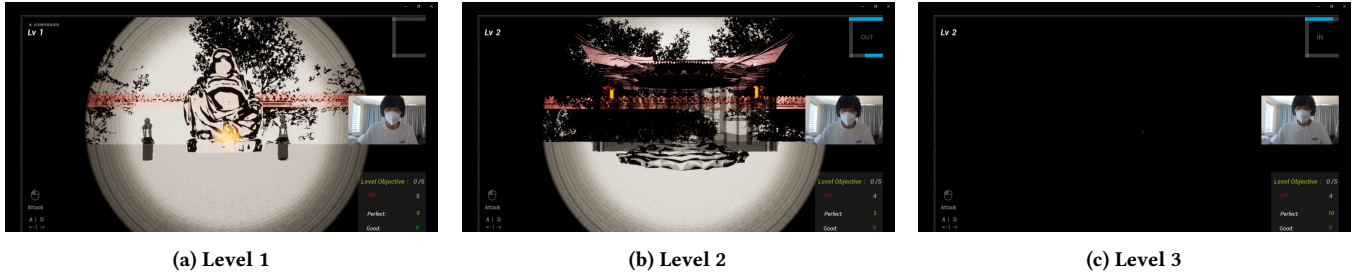


Figure 2: The Game Scenes



(a)



(b)

Figure 3: Breath Input System

This level-setting establishes a gradual learning curve, acclimatizing players to the unique auditory-centric gameplay.

3.2 Breath Input System

Listen to the Sword has an innovative breath input system. Breathing control is a key aspect of meditation and Chinese martial arts and spirituality, as it helps in calming the mind and concentrating energy. In *Listen to the Sword*, the Tu Na(Breath Control) system[6] is based on box breathing. Box breathing is a simple stress-relief technique involving a four-step process: inhale for four counts, hold for four counts, exhale for four counts, and hold again for four counts, which helps to calm the mind and relax the body. When players keep box breathing while playing the game the enemy's movements are slowed down, making it easier and less hurried for them. Otherwise, the players have to constantly move quickly to deal with enemies running towards them, causing significant stress.

To detect the player's breathing patterns, we developed a mask integrated with a sound sensor (see Fig 2) and used a modular, stackable development kit, the M5Stack [1]. The processed data from the M5Stack was then transmitted to a laptop for further analysis. This configuration allows for precise monitoring and recording of respiratory activity in a gaming context.

3.3 The Art Style

The art style of *Listen to the Sword* is heavily influenced by ancient Chinese aesthetics, creating a visually captivating and culturally immersive experience (See Fig 3). Drawing inspiration from traditional Chinese art forms and motifs, the game's art style combines elements of ink-wash painting and historical Chinese visuals. This art style embraces the visual aesthetics of ancient China, drawing from historical Chinese architecture, landscapes, and cultural elements, to create an environment that reflects the rich heritage of the region.

3.4 Expected Play Experience

Video games have been widely used and proved to be efficient in intercultural communication [2, 7, 9, 11, 13]. Due to high immersion and interactions, video games can enhance cultural and intercultural literacy, awareness, and skills, and influence perceptions and stereotypes about different geopolitical regions[13].

We presented our game at the Gotland Game Conference 2023 and collected some quick feedback during the exhibition. Many players noted a gradual increase in immersion as the visual cues within levels became less frequent:

"When I can only rely on auditory cues to identify enemies in the game, I find myself more immersed."

Another interesting feedback is, *"At the beginning of the game, I felt quite nervous, but as I consciously controlled my breathing, I gradually became calmer."* This tranquility reportedly enhanced their gaming experience. In addition to this, several players from non-Chinese cultural backgrounds, who had prior exposure to Wuxia movies, indicated that playing the game deepened their comprehension of the Wuxia genre.

4 CRITICAL REFLECTION

In *Listen to the Sword*, we wanted to create a traditional Chinese philosophical journey for players to learn the significant concept of Wuxia in a gamified method. Combining with unique audio-driven gameplay and a novel breath input system, we hope players could immersively experience the subtle and abstract idea, which could promote intercultural communication.

In the game design, we based our concepts on the idea of cultivation in Wuxia, gamifying the practices associated with Chinese Taoism and Zen Buddhism, including "Tu na" (breath control), "Zhi Guan" (stop and observe), and the martial arts state of "Wang Wo"

(selflessness). Coupled with educational content about these practices, our goal was to provide participants not only with an immersive experience of cultivation but also a deeper understanding of the Chinese cultural representation embodied in Wuxia. Based on the feedback we received at the Gotland Game Conference, we believe that the artistic style and scene design in the game can quickly evoke associations with Wuxia for players. Furthermore, as players become familiar with breath control and spatial audio cues during gameplay, not only is their immersion enhanced, but the concepts of "Tu Na" and "Zhi Guan" are better conveyed, ultimately leading to a deeper appreciation of the Chinese philosophy.

In previous games that used Wuxia as their core, the philosophical aspects of Wuxia were typically conveyed through text or visual cues. In contrast, in our game, we intertwine "Tu Na" and breath detection, "Zhi Guan" and spatial audio, as well as the temporal flow in the game to portray the state of "Wang Wo". By closely integrating these philosophical terms and in-game mechanisms related to Wuxia, participants can immersively experience the protagonist's exploration and achievements in the spiritual world during martial arts cultivation. We believe such an attempt can broaden the possibilities of games as a medium for cross-cultural communication.

In the future, we plan to conduct a proper user study to evaluate the effectiveness of our game in delivering Chinese culture, and we aim to propose design guidelines to support a deeper integration of cultural concepts with gaming. Also, we are planning to bring the game to VR platforms to further enhance its immersive experience.

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REFERENCES

- [1] Puput Dani Prasetyo Adi and Akio Kitagawa. 2019. A review of the Blockly programming on M5Stack board and MQTT based for programming education. In *2019 IEEE 11th International Conference on Engineering Education (ICEED)*. IEEE, 102–107.
- [2] Brian An, Donald Brown, and Stephanie Guerlain. 2019. The evaluation of a serious game to improve cross-cultural competence. *IEEE Transactions on Learning Technologies* 12, 3 (2019), 429–441.
- [3] Tianming Cao, Hongya Zhou, and Anqi Feng. 2023. Becoming a Xia: Constructing Cultural Identity in the History of Wuxia Games in China. *Games and Culture* (2023), 15554120231208735.
- [4] Robert L Chard. 1997. Transcendents, Sorcerers, and Women Warriors: Huanzhu-louzhu's Mountain Sword-warriors of Sichuan. *Chinoperl Papers* 20, 1 (1997), 169–195.
- [5] Thomas Cleary. 1997. *Stopping and Seeing: A Comprehensive Course in Buddhist Meditation*. Shambhala Publications.
- [6] Sanobar Dar, Aniko Ekart, and Ulysses Bernardet. 2022. The Virtual Human Breathing Coach. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 434–436.
- [7] Martin De Jong and Harald Warmelink. 2017. Oasistan: An intercultural role-playing simulation game to recognize cultural dimensions. *Simulation & Gaming* 48, 2 (2017), 178–198.
- [8] Tencent Games. 2023. Qu Zhong Jian - Android version. <https://www.taptap.cn/app/238345>
- [9] Gert Jan Hofstede and Paul Pedersen. 1999. Synthetic cultures: Intercultural learning through simulation games. *Simulation & Gaming* 30, 4 (1999), 415–440.
- [10] Leanne C McKay, Karl C Evans, Richard SJ Frackowiak, and Douglas R Corfield. 2003. Neural correlates of voluntary breathing in humans. *Journal of applied physiology* 95, 3 (2003), 1170–1178.
- [11] Vanessa Mills and Rebekah Smith. 2004. *Short-and long-term effects of participation in a cross-cultural simulation game on intercultural awareness*. DSTO Systems Sciences Laboratory.
- [12] Mirjana Prpa, Ekaterina R Stepanova, Thecla Schiphorst, Bernhard E Riecke, and Philippe Pasquier. 2020. Inhaling and exhaling: How technologies can perceptually extend our breath awareness. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [13] Elena Shliakhovchuk and Adolfo Muñoz García. 2020. Intercultural Perspective on Impact of Video Games on Players: Insights from a Systematic Review of Recent Literature. *Educational Sciences: Theory and Practice* 20, 1 (2020), 40–58.
- [14] Penelope Sweetser and Peta Wyeth. 2005. GameFlow: a model for evaluating player enjoyment in games. *Computers in Entertainment (CIE)* 3, 3 (2005), 3–3.
- [15] Stephen Teo. 2015. *Chinese martial arts cinema: The wuxia tradition*. Edinburgh University Press.
- [16] Stephen Teo. 2021. *Chinese martial arts film and the philosophy of action*. Routledge.
- [17] Iddo Yehoshua Wald, Amber Maimon, Lucas Keniger De Andrade Gensas, Noemi Guiot, Meshi Ben Oz, Benjamin W Corn, and Amir Amedi. 2023. Breathing based immersive interactions for enhanced agency and body awareness: a claustrophobia motivated study. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–7.
- [18] Joanneke Weerdmeester, Marieke MJW Van Rooij, and Isabela Granic. 2022. Visualization, self-efficacy, and locus of control in a virtual reality biofeedback video game for anxiety regulation. *Cyberpsychology, Behavior, and Social Networking* 25, 6 (2022), 360–368.
- [19] Peiheng Zhang. 1994. From Youxia to Wuxia: A Historical Inquiry into Chinese Xia Culture. *Fudan Journal: Humanities and Social Sciences* 3 (1994), 75–82.